

5 (a) (i) It is to increase the magnetic flux linkage between the coils **B1**

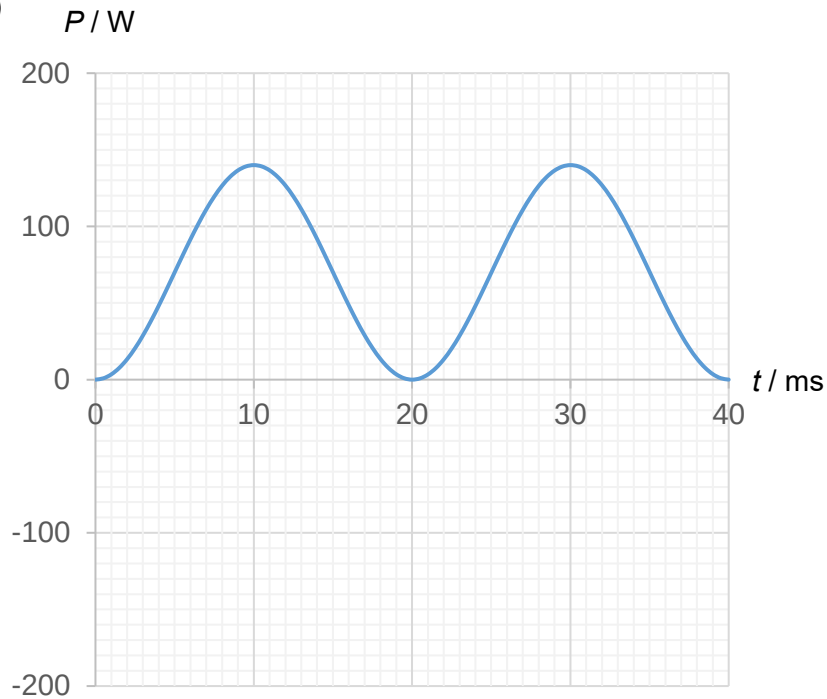
Comments: Most candidates answered correctly.

(ii) It is to reduce energy losses **B1**
by reducing induced currents **B1**

(b) (i) maximum $V_{OUT} = (N_s / N_p) V_p$
 $= (625 / 25\,000) 12\,000$
 $= 300\text{ V}$ **A1**

(ii) r.m.s. current = peak current $/\sqrt{2}$
 $= 300 / (640 \times \sqrt{2})$
 $= 0.33\text{ A}$ **A1**

(iii)



positive sinusoidal squared shape, with $P = 0$ at $t = 20, 40 \text{ ms}$
 and maximum P shown as 140 W
 two complete cycles, with period = 20 ms

B1**B1**

(c) For sinusoidal squared graph, mean power is half the peak power (140 W)

B1